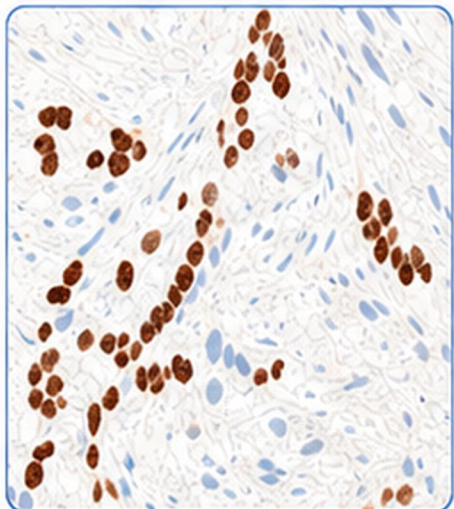


Input ER-IHC patch

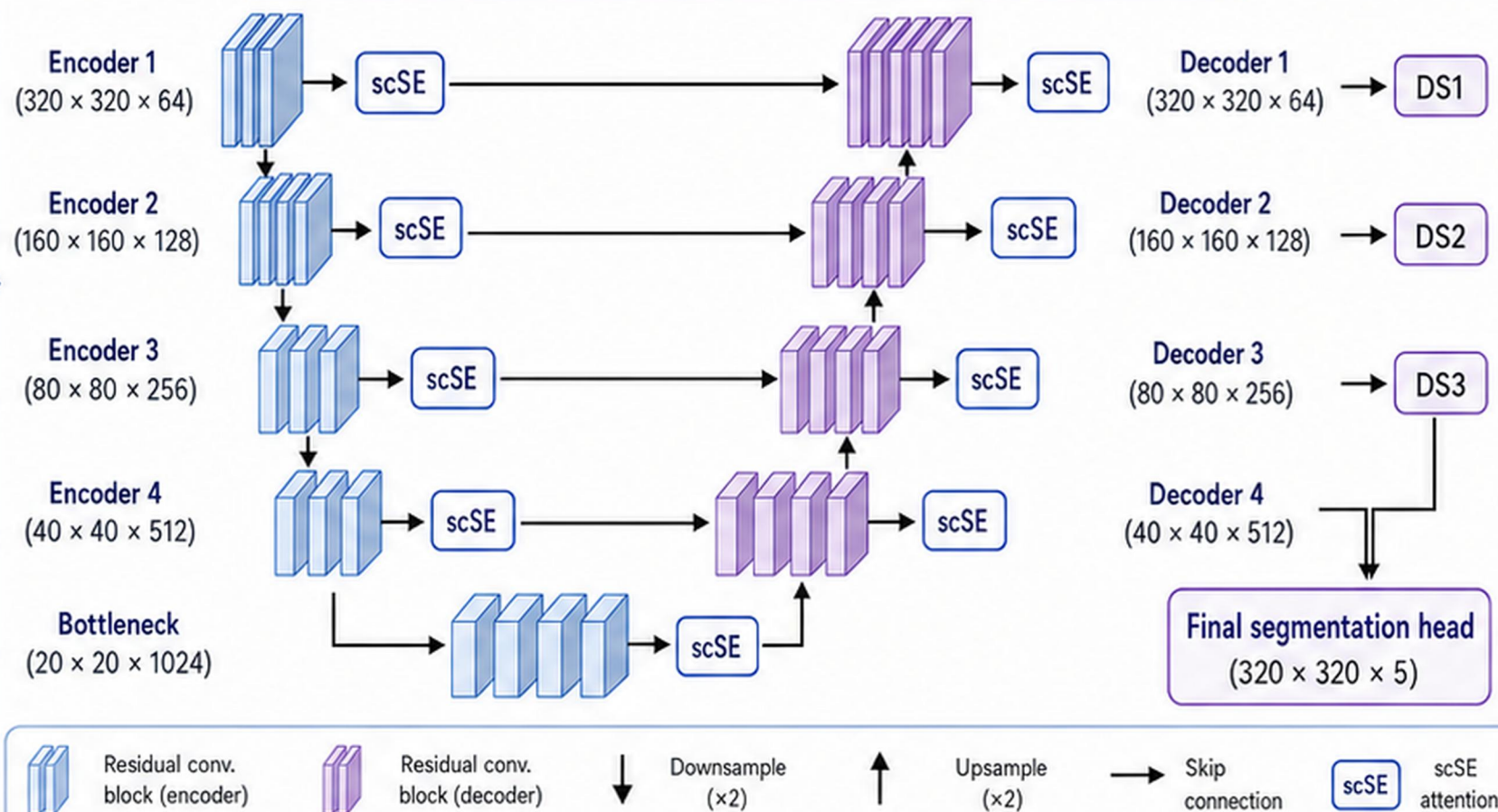
320 × 320 crop



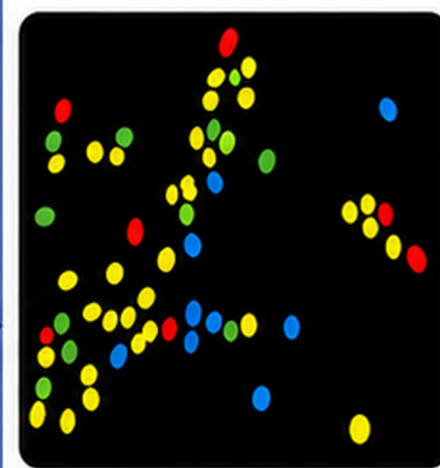
Training crop
sampled from
512 × 512
patch set

Proposed AMC-Ordinal architecture

ResUNet-DS backbone + scSE blocks + deep supervision

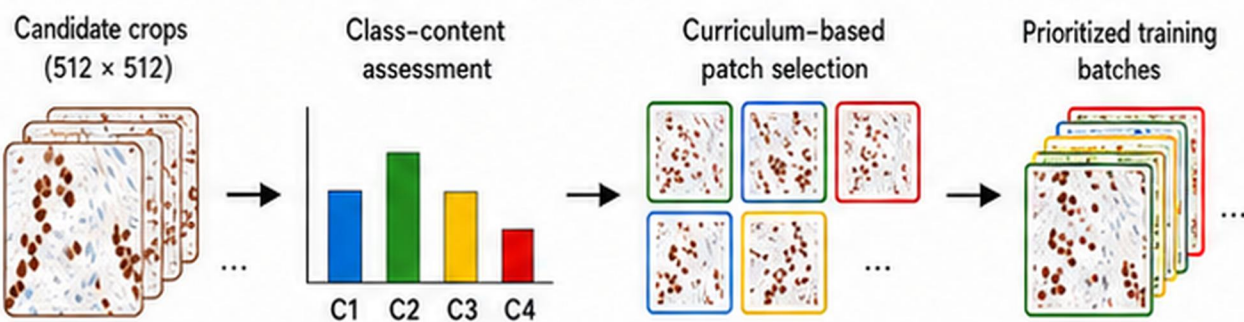


Predicted ordinal segmentation



- Background
- C1 (weak)
- C2 (low-moderate)
- C3 (moderate)
- C4 (strong)

Adaptive minority curriculum (AMC)



Dynamic emphasis on minority-rich regions
Improves learning for scarce ordinal classes (C2-C4)

DS1 DS2 DS3 Final output

Weighted CE + Dice

overall region
accuracy



Focal-Tversky loss

focus on hard /
minority regions



Foreground ordinal EMD loss

preserve ordinal
consistency among C1-C4



Joint optimization of all losses during training

- Minority-sensitive sampling
- Attention-enhanced backbone
- Deep supervision
- Ordinal-aware optimization



Targeted improvement for under-represented
C2 regions with competitive overall
segmentation performance.